

YIWEI WANG

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Education

Brown University

Expected May 2029

B.S. in Applied Mathematics-Computer Science, GPA: 4.00/4.00

Providence, RI

Relevant Coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Operations

Research: Probabilistic Models, Ordinary/Partial Differential Equations, Linear Optimization

Activities: Brown University Math Olympiad Problem Writer, Quant Trading at Brown Member

Honors and Awards

IMC Prosperity 4 Trading Competition Finalist
4× AIME Qualifier (Score: 12/15)
4× Los Angeles Math Field Day Gold Medalist
Coca-Cola Scholar Semifinalist (Top 1.27% of 105,000)

Skills

Languages: Python, C++, R, MATLAB, SQL
Libraries: NumPy, Pandas, XGBoost, Matplotlib
Technical: Algorithm Design, Time Series Analysis, Statistical Modeling, Optimization, Monte Carlo Simulation

Work Experience

Icarus Fund

June 2026 – Present

Quantitative Research Intern

New York City, NY

- Built **Python** cross sectional alpha research pipeline for **100+ S&P 500 signals**, computing **Rank IC, ICIR, and Newey West adjusted t-stats**; achieved **out of sample mean Rank IC of 0.053** with **59.7% positive days**.
- Designed **feature selection** pipeline using rank-correlation clustering, **VIF** multicollinearity filters, and **PCA**, identifying **top signals** including realized volatility, momentum, and volume factors from a universe of 50+ candidates.
- Prototyped **regime aware equity ranking model** combining **XGBoost** ranker (**500 estimators, cross validated grid search**) with **Hidden Markov Model** market state classification to adjust exposure across volatility regimes.

Research Experience

Hexagonal Tiling Combinatorics and Recurrence Relations

June 2023 – October 2023

Research Assistant | Advisor: Prof. Greg Dresden

Washington and Lee University

- Developed **recursive algorithms** to **enumerate tiling configurations** of hexagonal boards, extending **Fibonacci sequence theory** through mathematical modeling and analysis of trapezoidal tile arrangements.
- Derived novel **recurrence relations** integrating **Narayana's Cow sequence** to reduce computational complexity from **exponential to polynomial time**, improving the efficiency and scalability of large-scale combinatorial enumeration.
- Published integer sequence **A365352** in the *OEIS* and achieved a **top 8.9%** ranking among *Pioneer Research Journal* submissions, contributing original results in combinatorics research and advancing the study of recursive tiling sequences.

Project Experience

Volatility Dispersion Backtest | Python, NumPy, Pandas

June 2026 - Present

- Engineered **end to end dispersion trading framework** across **4 signal variants** (classic variance, replicated correlation, realized-vs-implied correlation, 0DTE gamma exposure) using **14 single-responsibility modules**; zero external backtest framework dependencies.
- Implemented **options math** from scratch including Black-Scholes **delta, gamma, vega, theta, variance swap replication** via log contract pricing, and **implied correlation back out**; validated against known correct synthetic fixtures with round trip pricing error less than **1e-9** and Greeks within **1e-7** of closed form truth.
- Built **6 automated validation gates** (look-ahead audit, cost sanity, reproducibility, VS-strike plausibility) enforcing strict **point in time data discipline**; designed pipeline to ingest options NBBO, vendor Greeks, and earnings data ahead of full historical backtest.

AlphaForge-Based Formulaic Alpha Mining Pipeline | Python, PyTorch, Pandas

May 2026 – Present

- Adapted **AlphaForge** GAN-based alpha mining framework to **US equities**, replacing Qlib/baostock pipeline with custom **local CSV loader** supporting **391 S&P 500 tickers** of daily OHLCV data spanning **2010–2026**.
- Mined **100-factor zoo** of formulaic alpha expressions via Generator/Predictor networks; achieved **out-of-sample mean daily IC of 0.051** and **ICIR of 0.55** on 2022 holdout across **long-short decile portfolio** of 391 stocks.
- Built **end-to-end backtesting infrastructure** including non-overlapping return correction for overlapping forward-return labels, SPY/equal-weighted benchmark comparison, and rolling **IC regime monitor** with kill-switch classification.

EMA Crossover Momentum Strategy | Python, Pandas, Matplotlib

May 2026 - June 2026

- Built **long/short EMA(30)/EMA(60) crossover** backtester on **10 years of 1-minute intraday AAPL** data resampled to **15-minute bars**; parameters selected via **grid search over 20 combinations**.
- Modelled **transaction costs per fill** (\$0.01 slippage + \$0.005 commission); strategy achieved **gross Sharpe of 1.38** (vs. buy and hold **0.84**) with **max drawdown of only 1.91%** (vs. **38.9%** for buy and hold) across **798 trades**.
- Designed **modular output pipeline** writing **6 structured CSVs** (trade book, EMA series, performance metrics, cost summary, baseline comparison) consumed by a **decoupled visualizer** producing equity curve and distribution charts.